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Two-dimensional Homotopy and Combinatorial Group Theory

Edited by

Cynthia Hog-Angeloni, Wolfgang Metzler
& Allan J. Sieradski

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University of Oregon

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Editors' Preface

It is well known that techniques developed for manifolds of higher dimensions don't suffice to treat open problems in dimensions three and four. The latter are inextricably tied to questions about the (simple-) homotopy type of 2-skeleta or -spines of these low-dimensional manifolds and, hence, to presentations of groups.

Basic work on two-dimensional homotopy dates back to K. Reidemeister and J.H.C. Whitehead. For instance, Whitehead gave an algebraic description of the homotopy type of 2-complexes. But, until the early 70's, one didn't have examples of 2-complexes with different homotopy type but equal fundamental groups and Euler characteristic. Since then considerable advances have been made, yielding, in particular, remarkable partial results on famous open problems like Whitehead's question, whether subcomplexes of aspherical 2-complexes are always aspherical themselves. The authors of this book have contributed to this development.

Because of its relations to decision problems in combinatorial group theory, two-dimensional homotopy probably will never take the shape of a complete theory. However, the occurrence of certain notions (e.g., the Reidemeister-Peiffer identities of presentations) in different questions is far from being accidental. The time has come to collect the present knowledge in order to stimulate further research.

This book contains the elements of both a textbook and a research monograph, and, hence, addresses students as well as specialists. Parts of the book have already been used to substantiate courses with concrete geometric and/or algebraic material. A student reader should know already some (algebraic) topology and algebra. We start with two introductory chapters on low-dimensional complexes. These are followed by chapters on prominent techniques including their applications to manifolds. Concluding chapters treat the present status of three famous conjectures (Whitehead, Zeeman, Andrews-Curtis). The coherent organization of the book includes cross references as well as a common index and an ample bibliography. But the chapters can also be read independently; they range from an introduction to

the specific topic(s) to a survey of latest results, with guidelines to the current literature. Particular emphasis is placed on covering open problems.

This book project was initiated by Wolfgang Metzler and presented to the majority of the authors at a workshop on Geometric Topology and Combinatorial Group Theory held in Luttach/Southern Tyrol (Italy) in August 1991. During this meeting, an approximate table of contents was developed.

The book demonstrates and documents mathematical cooperation. We mention, in particular, long-term activities of Micheal Dyer, Allan Sieradski and their former students, of Wolfgang Metzler and his former students and of Paul Latiolais' Fall Foliage Seminars, which have grown together in recent years. In addition to personal contact, modern electronic communication of (drafts of) sections or whole chapters was a basic ingredient in the production of manuscripts. This includes the illustrations that were based on sketches of the authors and drawn by Allan Sieradski using the Postscript drawing application Adobe Illustrator. The final layout was done by the editors.

All chapters were refereed twice, according to the usual textbook/journal standards, once by one or more member(s) of the team of authors, second by an external referee who had the option of remaining anonymous. As nobody chose to do so, we express our gratitude for their valuable service to: Juan Alonso (Stockholm), Stefan Bauer (Göttingen), Gerhard Burde (Frankfurt/Main), David Gillman (Los Angeles), Mauricio Gutierrez (Medford), Jens Harlander (Frankfurt/Main), Wolfgang Heil (Tallahassee), James Howie (Edinburgh), John Ratcliffe (Nashville), Nancy Waller (Portland) and Perrin Wright (Tallahassee). With their help, we have, in particular, tried to avoid mathematical and typographical errors. Our editorial efforts included eliminating conflicting use of terminology in different chapters; but we did not insist on standardized notations.

We invite all readers to communicate any remaining errors to us; we are also eager to learn about further progress on the mathematical substance of this book, which might be integrated into revisions of the text.

Last, but not least, we want to thank Roger Astley and David Tranah of Cambridge University Press for their continuous encouragement and advice throughout the project.

Cynthia Hog-Angeloni, Wolfgang Metzler, Allan J. Sieradski

Brombach (Taunus)

August 1993

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Chapter I

Geometric Aspects of Two-Dimensional Complexes

Cynthia Hog-Angeloni and Wolfgang Metzler

The aim of this introductory chapter is to provide a geometric background for the algebra and homotopy theory to follow. We also focus on geometric questions of intrinsic interest. The algebraic topology of later chapters is meant to contribute to their understanding and treatment. What we present is an extract of courses the authors have given on Low Dimensional Topology. These courses were enriched by selected topics from further chapters of this book and/or some of the material which this article only summarizes, together with references to other sources.

1 Complexes of Low Dimensions and Group Presentations

A crucial tool for dealing with geometric and homotopy theoretic problems in topology is the decomposition of certain spaces into disjoint unions of *cells* e_i^n of various dimensions, each (open) cell e_i^n as a subspace being homeomorphic to an open unit disc $\mathring{D}^n = \{x \in \mathbb{R}^n \mid |x| < 1\}$. Thus, we may obtain the structure of a *cell complex* K for the underlying topological space $|K|$. We denote by K^n (the *n-skeleton*) the subspace comprised of the cells of K of dimensions $\leq n$ with the induced cell structure. It is also standard terminology to indicate by the superscript n of K^n that K is a complex with

cells of at most dimension n (sometimes it is required that there are n -cells in an n -dimensional complex K).

The various notions of complexes differ by the conditions that are imposed on the closure $\bar{e}_i \subseteq |K|$ of a cell $e_i \in K$ and/or the *boundary* $\partial e_i = \bar{e}_i - e_i$. These conditions also regulate how the cells are “glued” together to yield the topology of $|K|$.

We assume that the reader is familiar with *simplicial complexes* (finite or infinite). Here each closed cell $\bar{e}_i$ is a union of cells and is homeomorphic to a simplex with all its faces by a homeomorphism which maps open cells to open cells. (A homeomorphism between cell-complexes which maps open cells to open cells is called *cellular* or an *isomorphism*.) As a “gluing condition” one mostly confines to the *weak topology* with respect to the $\bar{e}_i^n$, i.e.,

- (1) a subset of $|K|$ is closed iff its intersection with each $\bar{e}_i^n$ is closed.

Between 1939 and 1950, J.H.C. Whitehead published several papers which, on one hand, contain masterpieces of simplicial techniques. Some of these are basic for our geometric questions and will be cited in several sections of this article. On the other hand, Whitehead gradually was led to the insight that many of his results in homotopy theory actually hold for a generalization of simplicial complexes, where proofs and constructions can avoid a lot of hard work (checking the strong conditions of simplicial complexes and maps). In [Wh49₁], he introduced the notion of “CW-complexes,” for which the assumptions on the ∂e_i are far less restrictive than in the simplicial case. We give an inductive definition of *CW-complexes* which is particularly convenient in low dimensions. For the equivalence to Whitehead’s original definition, see Schubert ([Schu64], III. 3, Exercise 1 and Sieradski ([Si92], Chapter 15).

1.1 Inductive construction of *CW-complexes*

Definition: A *CW-complex* K is a space $|K|$ with a cell decomposition, whose skeleta are inductively constructed as follows:

- (a) K^0 is a discrete space, each point being a 0-cell.
- (b) K^n is obtained by attaching to K^{n-1} a disjoint family D_i^n of closed n -discs via continuous functions $\varphi_i : \partial D_i^{n-1} \rightarrow K^{n-1}$, i.e.: take the topological sum $K^{n-1} + \sum D_i^n$ and pass to the quotient space given by the identifications $x \sim \varphi_i(x)$, $x \in \partial D_i^n$. Each $\overset{\circ}{D}_i^n$ then projects homeomorphically to an n -cell e_i^n . φ_i is called an *attaching map* for e_i^n .

- (c) $|K| = \bigcup_{n=0}^{\infty} |K^n|$ is assigned the weak topology with respect to the $\bar{e}_i^n$ (as in (1)).

More generally, a cell complex is called a *CW*-complex, if it is isomorphic to one obtained by the preceding construction¹.

Remarks: (c) can be verified inductively for the skeleta (Exercise), hence it holds automatically if K is of finite dimension. The “*W*” in *CW* is motivated by Wweak topology, the “*C*” (Closure finite) by the fact that each $\bar{e}_i^n$ is contained in a finite union of cells (Exercise).

In contrast to the situation of simplicial complexes, $\bar{e}_i^n$ itself is not necessarily a union of cells; see Figure 1, where ∂e^2 is a point of e^1 , but not a 0-cell:

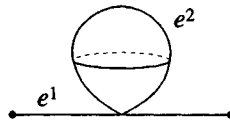


Figure I.1.

The following facts on *CW*-complexes can also be found in [Schu64] and [Si92]:

- (2) A *CW*-complex K is finite (i.e., consists of finitely many cells) iff $|K|$ is compact.
- (3) A covering space of a *CW*-complex K can be (uniquely) decomposed as a *CW*-complex $\tilde{K}$ such that the projection map $\tilde{K} \rightarrow K$ sends each cell $\tilde{e} \in \tilde{K}$ homeomorphically to a cell $e \in K$. (Note that the corresponding statement for closed cells is false in general, as the universal covering space of $S^1 = e^0 \cup e^1$ shows.) An attaching map for $\tilde{e}^n$ is obtained by appropriately lifting an attaching map $\varphi : \partial D^n \rightarrow K^{n-1}$ of e^n to $\tilde{K}^{n-1} = p^{-1}(K^{n-1})$.

We will deal mainly with finite *CW*-complexes and infinite ones that arise as covering complexes of (finite) complexes.

¹Attaching maps in this general case are those of an isomorphic model according to (a), (b), (c), composed with a cellular homeomorphism. We shall tacitly assume similar extensions of definitions to be made later on. Note that specific attaching maps are not considered data of the complex; compare § 2.1.

1.2 Questions of subdivision and triangulation

A 1-dimensional *CW*-complex is a graph. It may contain loops and more than one edge between two vertices; but by introducing new vertices, it can be subdivided² to become simplicial (details left as an exercise):



Figure I.2.

However, already in dimension 2 – due to the generality of attaching maps – there exist nontriangulable *CW*-complexes: Consider the finite(!) 2-complex K^2 of Figure 3, an *infinitely crumpled curtain* with three 0-cells, three 1-cells and one 2-cell, the attaching map of which oscillates on e_1^1 :

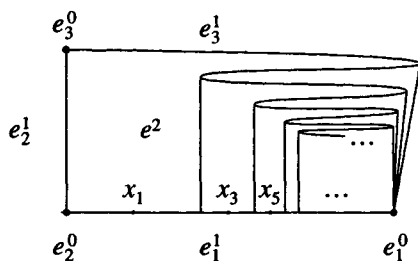


Figure I.3.

Note that all other marked points (e.g., the x_i) and lines of the drawing don't indicate further cells.

Theorem 1.1 *The CW -complex K^2 of Figure 3 is not homeomorphic to any simplicial complex.*

Proof: Triangulating appropriate neighbourhoods of the points x_i , $i = 1, 3, 5, 7, \dots$, one obtains the local homology groups $H_2(|K|, |K| - \{x_i\}) = \underbrace{\mathbb{Z} \times \dots \times \mathbb{Z}}_{(i-1) \text{ factors}}$. The number of “sheets” of the curtain coming together at x_i

²The general notion of subdivision for *CW*-complexes will be introduced below; see also footnote 5.

is thus seen to be an invariant of the local homeomorphism type of $|K|$ at x_i : $|K|$ has *infinitely many* distinct local homeomorphism types. Any triangulation of $|K|$ would have to be finite, as $|K|$ is compact (see (2)). But a finite simplicial complex has only *finitely many* local homeomorphism types, given by the stars of the simplices. Thus no such triangulation exists. $\square$

In the beginning of this century, the “Hauptvermutung” of combinatorial topology was raised, the question, whether homeomorphic simplicial complexes are combinatorially equivalent (i.e., if they become isomorphic after simplicial subdivisions). The terminology naturally generalizes to *CW*-complexes: A *subdivision* K' of a *CW*-complex K is a *CW*-complex K' with $|K'| = |K|$ and the property that each cell $e \in K$ is the union of certain cells $e'_i \in K'$. K and L are *combinatorially equivalent*, if they admit subdivisions K' (of K) and L' (of L) which are isomorphic. But the answers to the Hauptvermutung question are different in the simplicial and the *CW*-case:

After contributions of prominent mathematicians (e.g., Papakyriakopoulos, Moise, Bing, Milnor, Stallings, Kirby, Siebenmann), the answer to the simplicial Hauptvermutung is known to be “Yes” for (locally finite³ simplicial complexes of) dimensions ≤ 3 ; see [Br69], and “No” in dimensions ≥ 4 . (Closed 4-manifolds with exotic differentiable structures were first exhibited by M. Kreck [Kr84₂]; that such examples yield combinatorially distinct triangulations follows from [Mu60] together with [Ce68]).

But the Hauptvermutung for *CW*-complexes fails already in dimension 2, as the following example(s) will show:

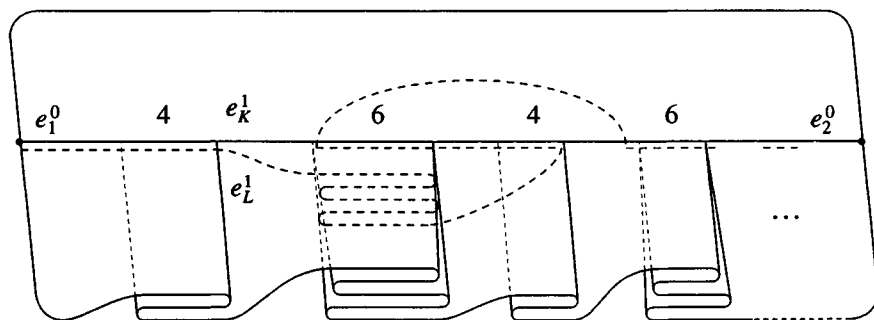


Figure I.4.

³A *CW*-complex is *locally finite*, if each point has a neighbourhood which meets only finitely many cells. (For simplicial complexes, this is equivalent to saying that the star of every simplex is finite (Exercise).) K is locally finite iff $|K|$ is locally compact; see [Schu64].

In Figure 4, K and L are finite CW -complexes with two 0-cells, three 1-cells and two 2-cells. They differ in their dissection of $|K| = |L|$ by the middle 1-cell e_K^1 resp. e_L^1 . The cell e_K^1 contains a sequence of (open) subintervals, where four or six sheets of 2-dimensional material meet, the sequence converging towards e_2^0 alternately: $4, 6, 4, 6, \dots$; e_L^1 is defined by the broken line and its periodic and shrinking continuation towards e_2^0 with the “rhythm” $4, 4, 6, 6, 4, 4, 6, 6, \dots$ of subintervals. It requires a little visualization to make sure that for the resulting 2-cells of L – above and below e_L^1 – none of the CW -conditions are violated.

Theorem 1.2 *K^2 and L^2 of Figure 4 are homeomorphic but combinatorially inequivalent.*

Proof: We use some easy facts on the sheets, which result from local homology considerations as in the proof of Theorem 1.2.

- a) Any subdivision K' of K , which is finite by (2), contains a 1-cell $e_{K'}^1$, (the one adjacent to e_2^0) characterized by the fact that almost all of the 4- and 6- sheeted subintervals are carried by it; L' analogously contains an $e_{L'}^1$.
- b) Any homeomorphism of $|K|$ to $|L| (= |K|)$ must map a 4- resp. 6-sheeted subinterval to a 4- resp. 6-sheeted subinterval. Remembering a), a cellular homeomorphism of K' to L' thus in particular would have to map $e_{K'}^1$ to $e_{L'}^1$.
- c) In e_K^1 – hence also in $e_{K'}^1$ – any two subintervals of type 4 are separated by a type 6 subinterval (and vice versa), whereas the different “rhythm” $4, 4, 6, 6, \dots$ in e_L^1 implies that $e_{L'}^1$ contains adjacent subintervals of one type without such a separation by one of the other type. The different rhythms thus contradict the monotony of a potential homeomorphism $e_{K'}^1$ to $e_{L'}^1$. Hence there is no cellular homeomorphism $K' \rightarrow L'$. $\square$

We refer to [Me67] for further subdivision phenomena, for instance:

- (4) The relation of combinatorial equivalence for CW -complexes is not transitive.

Geometric pathologies arising from general attaching maps don’t deserve too much interest of their own. But their occurrence either suggests the restriction to piecewise linear CW -complexes (see 1.4 and § 3 below) or it forces specific care in the proof of certain statements (see § 2.3 below).

1.3 Reading off presentations for π_1 of a CW -complex

The following material may be found in many textbooks. Proofs are based either on cellular/simplicial approximation (e.g., [Schu64]) or on the Seifert-van Kampen theorem (e.g. [Ma67], [Si92]); see also Chapter II, § 1 and § 2.

- (5) If K^1 is connected⁴, select a vertex e^0 as a basepoint and a *spanning tree*, i.e., a tree that consists of some edges and all vertices. Each remaining edge e_i^1 , together with an orientation, determines a closed path from e^0 as in Figure 5 (by connecting the initial and terminal vertex of e_i^1 with e^0 on the tree). The elements a_i of $\pi_1(|K^1|, e^0)$ given by these paths constitute a free basis of $\pi_1(|K^1|, e^0)$, i.e., $\pi_1(|K^1|, e^0)$ is the free group $F(a_i)$.

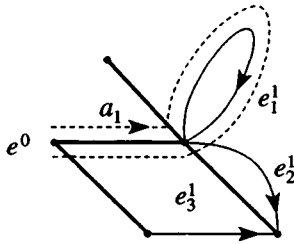


Figure I.5. Generating loops

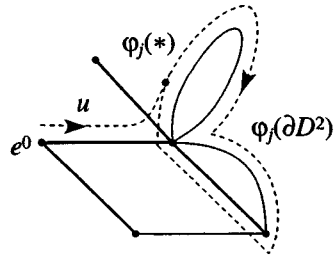


Figure I.6. Defining relations

- (6) If K^2 is connected, select a basepoint, a spanning tree and paths a_i for the 1-skeleton as in (5). For each 2-cell e_j^2 , an attaching map $\varphi_j : \partial D^2 \rightarrow K^1$, together with a base point $*$, an orientation of ∂D^2 and a connecting arc u from e^0 to $\varphi_j(*)$ (as in Figure 6) defines an element $R_j \in \pi_1(|K^1|, e_0)$ – a word R_j in the $a_i^{\pm 1}$ – which is trivial in $\pi_1(|K^2|, e^0)$. Moreover, $\pi_1(|K^1|, e^0) \rightarrow \pi_1(|K^2|, e^0)$ is surjective with the normal closure $N(R_j)$ (j ranging over all 2-cells) as the kernel.

Thus $\pi_1(|K^2|, e^0)$ has the *presentation* $\langle a_1, a_2, \dots | R_1, R_2, \dots \rangle$, i.e., the fundamental group $\pi_1(|K^2|, e^0)$ is given by *generators* a_i and *defining relations* R_j as the quotient $F(a_i)/N(R_j)$.

- (7) If K has K^2 as its 2-skeleton, then the natural map $\pi_1(|K^2|, e^0) \rightarrow \pi_1(|K|, e^0)$ is an isomorphism.

⁴A CW -complex is connected iff its 1-skeleton is pathwise connected (Exercise).

A crucial point of this chapter (and book) is that a finite, connected 2-complex not only determines a fundamental group, but – via (6) – a certain class of presentations, which doesn't contain all presentations of π_1 . Many homotopy invariants can be derived from any member of this presentation class. The first one to be mentioned is the Euler characteristic $\chi(|K|) = 1 - \#(a_i) + \#(R_j)$; see § 2.3.

As an example, note that for a compact, connected and simply connected K^2 the following statements are equivalent: (a) $|K|$ is *contractible* (i.e., $|K| \simeq *$); (b) $\chi(|K|) = 1$; (c) $H_2(|K|) = \{0\}$; (d) the presentations read off from $|K|$ according to (6) are *balanced*, i.e., the number of defining relations equals the number of generators (Exercise).

1.4 *PLCW-complexes*

In his papers [Wh39] and [Wh41₂], Whitehead didn't use general *CW*-complexes, but he confined his attention to specific ones which still admit simplicial⁵ subdivisions, so-called “membrane complexes”. They combine the advantage of a) providing complexes with “few cells” for a polyhedron with b) the piecewise linear (= p.l.) geometry, a main tool when embedding polyhedra into manifolds; see § 3. We refer to [Hu69], [RoSa72] and [Ze63-66] for the p.l. category, which is the proper home for simplicial theory when specific triangulations become irrelevant. In this category, membrane complexes are just what is given by the following definition:

Definition: A *PLCW-complex* is a locally finite *CW*-complex K together with a p.l. structure for $|K|$ such that

- (a) all closed cells and all skeleta are subpolyhedra;
- (b) K^n is obtained from K^{n-1} by a family of p.l. attaching maps $\varphi_i : \partial D_i^n \rightarrow K^{n-1}$ such that each subpolyhedron $e_i \cup K^{n-1}$ is p.l. homeomorphic, rel. K^{n-1} , to $D_i^n \cup_{\partial D_i^n} C(\varphi_i)$, where $C(\varphi_i)$ is the p.l. mapping cylinder of φ_i .

Here D^n is meant to be equipped with the p.l. structure given by a fixed homeomorphism of D^n to the n -cube I^n . b) involves results on p.l. mapping cylinders (existence, triangulations) which date back to [Wh39]; see the

⁵A *simplicial subdivision* K' of a *CW*-complex K is a subdivision of K by a simplicial complex K' .

discussion and the references in ([CoMeSa85], § 2). They yield, in particular, that by attaching finitely many n -cells to a $PLCW$ -complex K^{n-1} via p.l. attaching maps, the result naturally becomes a $PLCW$ -complex. Hence there is an inductive construction yielding finite $PLCW$ -complexes. Exercise: Derive concrete triangulations in the 2-dimensional cases (8), (9) which follow.

Some classes of finite 2-dimensional $PLCW$ -complexes:

- (8) *Reidemeister complexes* (see [Re32]): For the construction of K^2 from K^1 , let each edge of K^1 be equipped with a linear structure; each ∂D_j^2 is subdivided as a polygon; $\varphi_j : \partial D_j^2 \rightarrow K^1$ maps each edge linearly onto an edge of K^1 or onto a vertex. φ_j thus defines a closed *edge path* in K^1 . The notion of a Reidemeister complex agrees with the one of a combinatorial CW -complex of dimension 2; see Chapter II, § 1.2.

After subdividing K^1 , any (finite) $PLCW$ -complex becomes a Reidemeister complex (Exercise). Simplicial 2-complexes are Reidemeister complexes in which each triangle of a simplicial K^1 may be filled in at most once.

- (9) *Standard complexes of (finite) presentations*: As an “inverse” process to 1.3, we may associate to a finite presentation $\mathcal{P} = \langle a_1, \dots, a_g | R_1, \dots, R_h \rangle$ a Reidemeister-complex $K_{\mathcal{P}}$ with one vertex. Its 1-skeleton is a bouquet of circles with an oriented e_i^1 for each a_i :

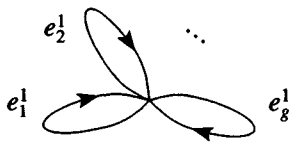


Figure I.7.

The 2-cells e_j^2 correspond bijectively to the R_j , which determine closed edge paths φ_j as the corresponding attaching maps.

Note that the sequence $R_1, \dots, R_h$ may contain repetitions and/or the trivial word $R_j = 1 \in F(a_i)$. Those relators nevertheless must *not* be suppressed when constructing the standard complex $K_{\mathcal{P}}$, which hence may contain different 2-cells with the same boundary and/or closed 2-cells that are 2-spheres.

There is no general rule as to whether the R_j are assumed to be (cyclically) reduced or whether R_j may contain adjacent inverse letters a_i, a_i^{-1} . Reducing

such “spurs” in the φ_j does not affect the (simple-)homotopy type of $K_{\mathcal{P}}$, see Lemma 2.1 of § 2.1 below; but it may drastically change the embedding behavior of $K_{\mathcal{P}}$ into 4-space, see § 3.2 below.

Amongst the standard complexes are those for canonical dissections of closed 2-manifolds, ([Schu64], III 5.8; [Si92], Chapter 13), but in general more than two local sheets may meet at an e^1 . It is nevertheless possible to apply certain moves to finite 2-dimensional CW -complexes which reduce the local complexity by achieving general position; see Ikeda [Ik71], Wright [Wr73], [Wr77] and Remark 1 after the proof of Theorem 3.1 in § 3.1. The result is a compact 2-dimensional polyhedron in which each point has a star of one of the following p.l. types:

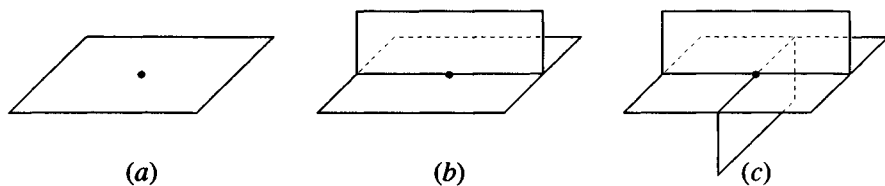


Figure I.8.

Such a polyhedron is called a *closed fake surface*; its *intrinsic 2-skeleton* consists of points of type (a); type (b) resp. (c) defines the *intrinsic 1-resp. 0-skeleton*⁶. As closed 2-manifolds show, these intrinsic skeleta in general don't dissect a closed fake surface into a cell complex. This additional requirement gives rise to the following definition:

- (10) A *special* (or: *standard* polyhedron) is a closed fake surface $|K|$ with the property that the components of the intrinsic skeleta are the cells of a $PLCW$ -decomposition K (for the given p.l. structure) of $|K|$.

Standard polyhedra are also Reidemeister complexes (Exercise).

In the literature, one sometimes has to deduce the precise meaning of “2-complex” from the context; and if specific names are given, their meaning is not always the same.

⁶Casler [Ca65] and Wright [Wr77] define the intrinsic 1-skeleton to consist of the points of type (b) or (c); consequently, these authors don't introduce the separate notion of the intrinsic 2-skeleton. Compare also the notions of Chapter XI, § 5.1.

2 Simple-Homotopy and Low Dimensions

2.1 A survey on geometric simple-homotopy

Throughout this section, all spaces are assumed to be compact.

In the definition of a *CW*-complex in order to attach an n -cell e^n , we had to form the quotient map $p : K^{n-1} + D^n \twoheadrightarrow K^{n-1} \cup_{\varphi} D^n$ that identifies each $x \in \partial D^n$ with $\varphi(x)$, where $\varphi : \partial D^n \rightarrow K^{n-1}$ is the attaching map for e^n . The restriction of p to D^n yields a characteristic map for e^n :

Definition: A map $\Phi : D^n \rightarrow K$ is *characteristic* for an n -cell e^n of K if Φ maps the interior $\overset{\circ}{D}^n$ homeomorphically onto e^n and $\Phi|_{\partial D^n}$ is an attaching map.

Note that attaching maps and characteristic maps are by no means unique. In the case of characteristic maps, an easy example is obtained by a self-homeomorphism of I^n on the one hand and shrinking $I^{n-1} \times \{1\}$ in I^n to a point (while mapping the interior homeomorphically) on the other hand. Nevertheless, it is true that

- (11a) Exercise: Two attaching maps for an n -cell e^n are homotopic up to composition with an orientation reversing map of ∂D^n to itself.
- (11b) Exercise: Two characteristic maps for e^n determine homotopic maps $(D^n, \partial D^n) \rightarrow (K^n, K^{n-1})$ up to composition with an orientation reversing homeomorphism of D^n to itself.

We now want to describe particularly simple operations which do not change the homotopy type of a given complex L , namely the attaching of an n -ball along an $(n-1)$ -ball in its boundary: (D^n, D^{n-1}) are assumed to form a *standard ball pair*, i.e., a pair homeomorphic to $(I^{n-1} \times I, I^{n-1} \times \{1\})$. Let $\varphi : D^{n-1} \rightarrow L^{n-1}$ and define $|K|$ to be $|L| \cup_{\varphi} D^n$. Then $|K|$ has a natural cell decomposition consisting of cells of L together with new e^n and e^{n-1} , and the associated characteristic map $\Phi : D^n \rightarrow K$ fulfills

$$(12) \quad \Phi|_{D^{n-1}} \text{ maps to the subcomplex } L^{n-1} \text{ and}^7$$

$$(13) \quad \Phi|_{\overline{\partial D^n - D^{n-1}}} \text{ is characteristic for } e^{n-1}.$$

⁷Equivalently, the roles of D^{n-1} and $\overline{\partial D^n - D^{n-1}}$ could be interchanged simultaneously in (12) and (13) which seems more adequate in some applications; see also Figure 9.

This gives rise to the following

Definition: An *elementary collapse* (of dimension n) $K \xrightarrow{e} L$ is the transition from K to L if L is a subcomplex⁸ of K , $K = L \cup e^n \cup e^{n-1}$ where $e^n, e^{n-1} \notin L$ such that there exists a standard ball pair (D^n, D^{n-1}) and a characteristic map $\Phi : D^n \rightarrow K$ for e^n fulfilling (12) and (13)⁹. The inverse operation $L \xrightarrow{e} K$ is called an *elementary expansion* (of dimension n):

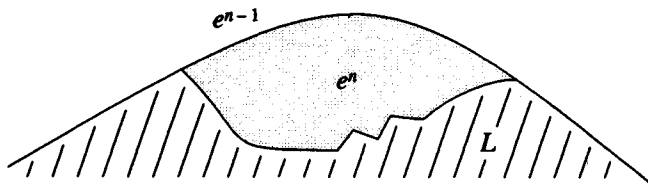


Figure I.9.

Note that for $K \xrightarrow{e} L$ (resp. $K \xrightarrow{e} L$) there exists a (strong) deformation retraction (resp. the inclusion) from $|K|$ to $|L|$. A sequence $K_0 \xrightarrow{e} \dots \xrightarrow{e} K_m$ is called a *collapse* $K_0 \searrow K_m$ resp. an *expansion* $K_m \nearrow K_0$. K_m is called a *spine* of K_0 . A sequence where both, collapses and expansions, are allowed, is abbreviated by $K_0 \xrightarrow{\sim} K_m$. $|K_0|$ and $|K_m|$ are then related by a *deformation* defined to be a composition of retractions and inclusions as above. Sometimes the maximal dimension of involved cells will be added in the notation and we shall speak of an n -(dimensional) *deformation* $K_0 \xrightarrow{n} K_m$. Finally, we define a *simple-homotopy equivalence* (in short: *sh-equivalence*) to be a map which is homotopic to a deformation. This concept is due to J. H. C. Whitehead [Wh50]; we also refer to [Co73].

The definition of an elementary collapse makes sense not only in the *CW*-setting. For *polyhedra* K and L , D^n is attached along a p.l. embedding φ ; hence $|K| - |L|$ is a p.l. ball D^n , which has $|L| \cap D^n$ as a face; compare ([RoSa72], Chapter III). If K and L are *PLCW-complexes*, a *CW-collapse* as in the above definition is called a *PLCW-collapse* and induces a p.l. collapse $|K| \searrow |L|$; see ([CoMeSa85], Remark 4). Finally, if K and L are *simplicial*

⁸A subcomplex of a *CW*-complex K is a subset L of cells which constitutes a *CW*-structure for the subspace $|L| = \bigcup e_i$, $e_i \in L$. This holds iff $\bar{e}_i \subseteq |L|$ for all $e_i \in L$. The skeleta of a *CW*-complex are examples of subcomplexes.

⁹ e^{n-1} is then called a *free face* of K . If $e^n, e^{n-1} \in K$ and $K^{n-1} \cup e^n \xrightarrow{e} K^{n-1} - e^{n-1}$, then e^{n-1} is a *free face* of e^n . A free face of K is always a free face of the corresponding e^n .

complexes, e^n and e^{n-1} are supposed to be (open) simplices; see [Hu69]. In all these settings, the basic terms can be defined analogously and lead to a simple-homotopy theory. The latter though, roughly speaking, turns out always to be the same; see Proposition 2.3 and (18) below for a precise meaning when comparing the *CW* with the *PLCW* setting; for *CW* versus polyhedra, we refer to ([RoSa72], Appendix B5); polyhedra versus simplicial complexes see ([Hu69], Chapter II). To avoid confusion, we shall sometimes put a superscript CW , $p.l.$, $PLCW$, s on the arrow to indicate the appropriate setting.

Returning to *CW*-complexes, we note that

- (14) (Exercise:) the elementary steps in a deformation can be reordered (up to isomorphism¹⁰ of the resulting complex) so that first all expansions are done in increasing dimensions and then all collapses in decreasing dimensions.

An important application of deformations arises when deforming the attaching map of an n -cell to some n -complex:

Lemma 2.1 *Let K be an n -dimensional complex and let $\varphi \simeq \psi : S^{n-1} \rightarrow K$ with $\varphi(S^{n-1}), \psi(S^{n-1}) \subseteq K^{n-1}$ be attaching maps for n -cells e_φ^n resp. e_ψ^n . Set $K_\varphi := K \cup_\varphi D^n$ and $K_\psi := K \cup_\psi D^n$. Then $K_\varphi \xrightarrow{n+1} K_\psi$.*

Proof: Attach $S^{n-1} \times I$ in the boundary of an $(n+1)$ -ball $D^n \times I$ via the given homotopy $H : S^{n-1} \times I \rightarrow K$

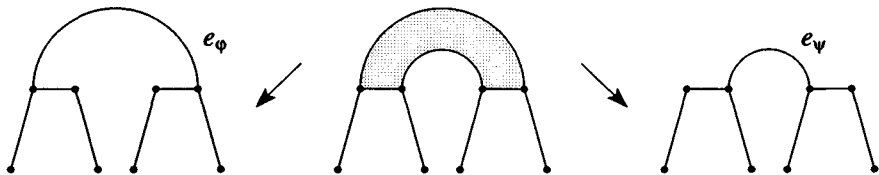


Figure I.10.

and let $K^{n+1} = K \cup_H (D^n \times I)$ be the resulting space with its naturally induced cell decomposition. It has a single $(n+1)$ -cell with two free faces e_φ

¹⁰The original deformation may remove and reintroduce certain material which, in the resulting one, has to be formally separated.

and e_ψ . Then $K_\varphi \xrightarrow{e} K^{n+1} \xrightarrow{e} K_\psi$ by a collapse through the top e_φ resp. the bottom e_ψ of the $(n+1)$ -cell. $\square$

Here, only one $(n+1)$ -cell is expanded to a CW -complex and collapsed immediately afterwards. Such a deformation is called an $(n+1)$ -dimensional *exchange* or *transient move*.

If we wish to do homotopies on attaching maps for cells of dimensions lower than the one of the complex, the following lemma (on simple-homotopy extension) is useful:

Lemma 2.2 *Let K^n be a finite CW -complex with m -skeleton K^m , $m < n$. Then for a given deformation $K^m \xrightarrow{m+1} L^m$ there exists a finite CW -complex L^n with m -skeleton L^m such that $K^n \xrightarrow{n+1} L^n$ holds.*

Proof: $K^m \xrightarrow{m+1} L^m$ yields a deformation $K^m \nearrow P^{m+1} \searrow L'^m$ such that $\alpha)$ L'^m is isomorphic to L^m and $\beta)$ the $(m+1)$ -cells of P^{m+1} can also be expanded to K^{m+1} to result in Q^{m+1} ; compare (14). But in general the collapse corresponding to $P^{m+1} \searrow L'^m$ cannot be performed in Q^{m+1} . Instead, using the strong deformation retraction induced by the collapse $P^{m+1} \searrow L'^m$, re-attach the $(m+1)$ -cells of K^{m+1} differently to P^{m+1} so that they avoid the free m -faces of that collapse¹¹. As the preceding lemma shows, the resulting complex Q'^{m+1} can be obtained from Q^{m+1} by an $(m+2)$ -deformation; and afterwards the collapse $P^{m+1} \searrow L'^m$ can be performed in Q'^{m+1} . In total, we have constructed the sequence $K^{m+1} \nearrow Q^{m+1} \xrightarrow{m+2} Q'^{m+1} \searrow L'^{m+1}$, where L'^{m+1} is the union of L'^m and the re-attached $(m+1)$ -cells of K^{m+1} . A further $(m+2)$ -deformation of L'^{m+1} to an L^{m+1} with m -skeleton L^m then follows from disjointness considerations; compare [Co73], 5.2.b). The proof is completed by an inductive application of this step. $\square$

The two lemmas together yield the

Proposition 2.3 *Every n -dimensional finite CW -complex $(n+1)$ -deforms to a $PLCW$ -complex.*

Proof: Assume that the m -skeleton of a given finite CW -complex K is already $PLCW$, which it is for $m = 0$. As long as $m < n$, deform the attaching maps of the $(m+1)$ -cells to become p.l.. Let L^{m+1} be the $PLCW$ -complex

¹¹ Attaching maps for the new $(m+1)$ -cells are thus given by those of K^{m+1} composed with the deformation.

obtained by attaching the $(m+1)$ -cells via these p.l. maps to K^m . By Lemma 2.1, K^{m+1} $(m+2)$ -deforms to L^{m+1} ; by Lemma 2.2 this induces a deformation $K \xrightarrow{n+1} L$, where the $(m+1)$ -skeleton L^{m+1} of L now is a *PLCW*-complex. $\square$

- (15) Similarly, we can apply an $(n+1)$ -deformation to standardize the 1-skeleton of any finite connected *CW*-complex K^n to a bouquet of circles (see § 1.4, Figure 7):

Attaching maps for 1-cells that are not in the spanning tree are always homotopic to the constant map to the basepoint, and when they are all deformed away from the tree (except for the basepoint), the tree can be collapsed. The two lemmas again convert this procedure into an $(n+1)$ -dimensional deformation. $\square$

Each choice of p.l. approximations for attaching maps during the inductive construction applied to K in the proof of Proposition 2.3, gives rise to a *PLCW*-complex which will be called a *PLCW-approximation* of K .

- (16) Exercise: Any two *PLCW*-approximations P, Q of a finite n -dimensional *CW*-complex K fulfill $P \xrightarrow{n+1, PLCW} Q$.
- (17) Exercise: Let $K \xrightarrow{e} L$ where K and L are finite *CW*-complexes. Then there are *PLCW*-approximations P resp. Q for K resp. L for which $P \xrightarrow{e, PLCW} Q$ holds.

These two facts together imply that

- (18) finite *PLCW*-complexes P, Q which are simple-homotopy equivalent in the *CW*-sense, are related by a deformation where all intermediate steps are *PLCW*:

In a deformation from P to Q that uses intermediate stages K_i , choose *PLCW*-approximations P_i for each K_i . By (17), there are *PLCW*-approximations P'_i, P'_{i+1} for K_i and K_{i+1} such that P'_i and P'_{i+1} are related by an elementary collapse in the *PLCW*-sense. But then by (16), also P_i and P_{i+1} are related by a deformation where all intermediate steps are *PLCW*. $\square$

Note that this procedure will in general raise the dimension of a deformation by 1.

In § 2.3 (Remark 2 after the proof of Theorem 2.4), we shall see that 2-dimensional *PLCW*-complexes which 3-deform into each other in the *CW*-sense, actually are related by a 3-dimensional *PLCW*-deformation.

The fundamental questions of simple-homotopy theory are, whether every homotopy equivalence is simple or (if not) whether homotopy equivalent complexes are always simple-homotopy equivalent. The answer to these questions turns out to be different according to the dimension of the complexes:

- (19) In dimension 1, using Nielsen's theorem [Ni19] on the decomposition of automorphisms of free groups, we shall see in § 4.1 that in fact every continuous map between finite, connected 1-complexes which induces an isomorphism of fundamental groups is homotopic to a deformation of dimension 2.
- (20) For all dimensions ≥ 3 , there exist finite, connected complexes which are homotopy equivalent but simple-homotopy distinct; see [Co73], 24.4.

(20) needs the algebraic theory of simple-homotopy, in particular the *Whitehead group* $Wh(\pi)$ of a group π ; see [Wh50] and [Co73]. The first examples arose in the p.l. classification of certain closed 3-manifolds, the lens spaces (see § 3.1) and their higher dimensional analogues (Reidemeister [Re32] and Franz [Fr35]).

In dimension 2, the one of our main interest, an additional difficulty is the realization of the algebraic topological construction by different presentations of the same group; compare § 4.4. Nevertheless

- (21) there exist finite, connected 2-complexes which are homotopy equivalent but simple-homotopy distinct (Metzler [Me90] and Lustig [Lu91₁]); see Chapter VII, Theorem 3.2.

A related question is, whether finite, homeomorphic *CW*-complexes are of the same simple-homotopy type. In the p.l. case for dimensions ≤ 3 , this is a consequence of the Hauptvermutung (compare § 1.2) and the fact that

- (22) a subdivision of a *CW*-complex can be realized by a deformation (Theorem 12 of [Wh50] and [Co73], 25.1).

But the topological invariance of simple-homotopy type holds for arbitrary dimensions and *CW*-complexes¹², which is a consequence of a famous *Theorem of Th. Chapman* [Ch74] and *J. West* [We71]; see also [BrCo74]:

¹²For dimension 2, this follows also from § 2.3, Theorem 2.7.

- (23) If K and L are finite CW -complexes (resp. polyhedra) then $f : K \rightarrow L$ is a simple-homotopy equivalence if and only if $f \times Id_Q : K \times Q \rightarrow L \times Q$ is homotopic to a homeomorphism.

In (23), Q denotes the Hilbert cube $\prod_{j=1}^{\infty} I_j$.

On the other hand, *collapsibility*¹³ (even of $PLCW$ -complexes) is not a topological invariant as long as subdividing is not allowed: Goodrick [Go68] has given examples of triangulations of the topological n -ball for $n \geq 3$ which are not simplicially collapsible. However, by Zeeman ([Ze63-66], Chapters I and III), it follows that, for any triangulation of a topological n -ball, there is an integer k such that the k -th barycentric subdivision of the triangulation is simplicially collapsible. A result of Chillingworth ([Ch67], [Ch80]) shows that simplicial collapsibility of complexes of dimension at most 3 carries over to subdivisions. Together with the Hauptvermutung, this implies that a $PLCW$ -complex that has dimension at most 3 and is topologically homeomorphic to a collapsible $PLCW$ -complex becomes (simplicially) collapsible after subdivision.

2.2 Some examples

Every compact, connected n -dimensional p.l. manifold M with nonempty boundary collapses to a spine of dimension $\leq n - 1$: Each (open) n -simplex is accessible from ∂M by a path which misses the $(n - 2)$ -skeleton. Thus all n -simplices can be removed by collapses:

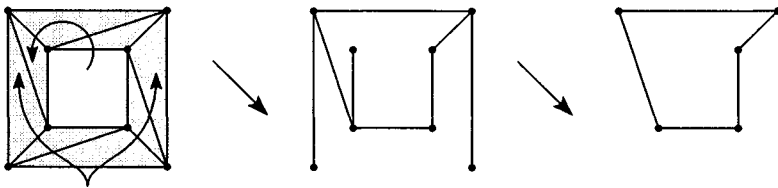


Figure I.11.

In some cases it may be necessary to do some expansions before getting down to a “small” complex. In particular, contractible complexes K do not necessarily collapse to a point.

¹³ K is called *collapsible*, i.e., $K \searrow *$, if K has a spine which is a point.

A famous example for this is the *dunce hat* H introduced by E. C. Zeeman [Ze64₁]. It is the standard complex of the presentation $\langle a|aaa^{-1} \rangle$:

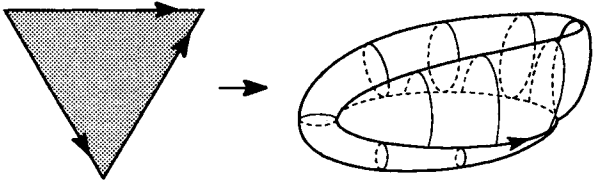


Figure I.12. Dunce hat construction

The dunce hat doesn't allow any collapse, as each 1-cell locally bounds at least two sheets. By use of local homology as in § 1, this “can't start”-argument generalizes to any CW -decomposition of $|H|$ and to the p.l. setting. Nevertheless we will establish p.l. collapses

$$(24) \qquad |H| \searrow |H| \times I \searrow *$$

via suitable $PLCW$ -decompositions and -collapses¹⁴ of $|H| \times I$:

The collapse $(|H| \times I \searrow |H|)$ is obvious: $|H| \times I$ inherits a natural “cylindrical” structure $H \times I$ from H which can be collapsed with decreasing dimensions through the “tops” $e^n \times \{1\}$ down to the “bottom” $H \times \{0\}$. For the collapse $(|H| \times I \searrow *)$ we subdivide $H \times I$ by a “slanted sheet” as in Figure 13 and collapse the 3-dimensional material from above and below; see Figure 14:

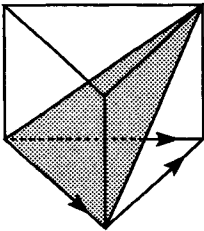


Figure I.13.

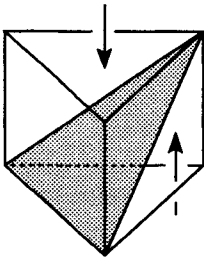


Figure I.14

What is left, is the edge a of H , crossed with I , together with the sheet, the slanted edges of which are identified:

¹⁴Later on we will be less strict with the notation, when p.l. collapses are applied to a $PLCW$ -complex.

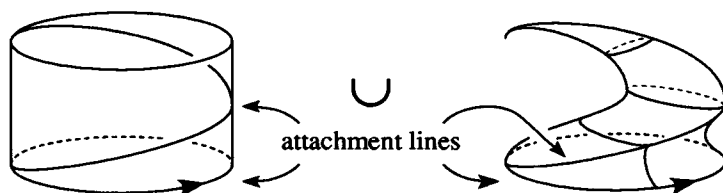


Figure I.15.

If in a next step we begin by collapsing through $a \times \{1\}$

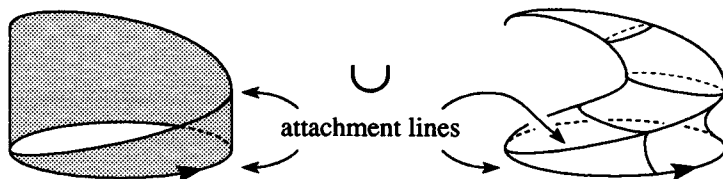


Figure I.16.

and then through $(\{\text{base point}\} \times I)$, we get rid of all 2-dimensional material of $a \times I$. At this point, only the sheet is left over and can easily be collapsed to the basepoint of $H \times \{0\}$. $\square$

Even though the dunce hat embeds into 3-space, it is not quite easy to visualize. *Bing's house*, which is another example of a contractible but non-collapsible complex, naturally occurs as a spine of the 3-ball:

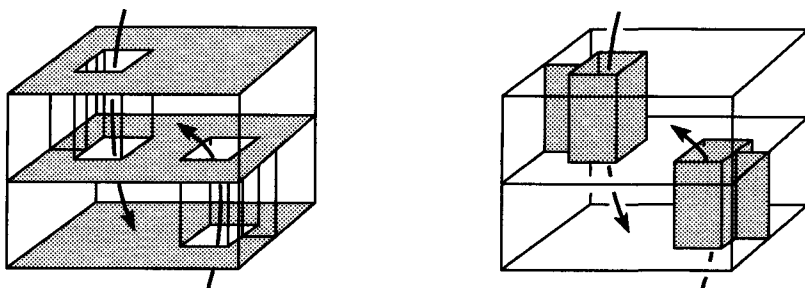


Figure I.17. Floors and panels of Bing's house; enter the bottom room from the roof and the top room from below.

We have given a modified version¹⁵ of Bing's house which has the further advantage of being a special polyhedron.

The same “can't start”-argument as for the dunce hat implies the non-collapsibility, while $\{\text{Bing's house}\} \searrow I^3 \searrow *$ yields the contractibility. Note that we have started with something obviously collapsible, namely the 3-ball, but one of the collapsing strategies “got stuck”: we ended up with a 2-complex that doesn't have any free faces. One dimension lower though *in order to check the collapsibility of a complex K^2 , we can just do collapses “helter-skelter” (R. Lickorish) and see whether eventually we come down to a point.* To see this, observe first that $K^n \searrow *$ implies that no collapsing strategy gets stuck prior to the collapse of all of the highest dimensional material. (A strategy that hasn't removed all n -cells yet, could be extended by the first collapse of the remaining n -cells in the original strategy.) In particular, when dealing with a collapsible 2-complex, any sequence of collapses can be continued until a graph is obtained. But a graph is contractible if and only if it is collapsible namely if and only if it is a tree. This completes the argument. $\square$

2.3 3-deformation types and Q^{**} -transformations

The process in § 1.3 to read off a presentation $\mathcal{P} = \langle a_1, \dots, a_g | R_1, \dots, R_h \rangle$ for the fundamental group of a finite, connected CW -complex K^2 involves the choice of a spanning tree, base points, orientations, connecting arcs and attaching maps. But different choices only amount to

- (25) a) replacing an R_j by $wR_j^{\pm 1}w^{-1}$ for some $w \in F(a_i)$ (compare Exercise (11a) above on different attaching maps for a 2-cell e_j^2), and
- b) replacing the a_i (in the R_j) by the result of a change of basis¹⁶ or of an isomorphism applied to $F(a_i)$.

These modifications of $\mathcal{P}$ are among those that can be achieved using the elementary transformations (26), (27), (28), which preserve (the isomorphism type of) the group $\pi = F(a_i)/N(R_j)$:

- (26) a) $R_j \rightarrow wR_jw^{-1}$ for some j and $w \in F(a_i)$ (*conjugation*),
- b) $R_j \rightarrow R_j^{-1}$ for some j (*inversion*), and

¹⁵See [Co73] for the original design.

¹⁶If T and T' are two different spanning trees for K^1 , then one can add to T an edge e^1 in $T' - T$ and remove one of the $T - T'$ edges of the path which connects the distinct endpoints of e^1 in T . Finitely many such changes convert T into T' . They yield concrete elementary Nielsen transformations (compare (27)) which change the basis a_i of $\pi_1(|K^1|, e^0)$ with respect to T into a basis with respect to T' ; see [Re32].

- c) $R_j \rightarrow R_j \cdot R_k$ or $R_k \cdot R_j$ for some $k \neq j$ (multiplication on right or left).

In each elementary step, the relators with an index $\neq j$ remain unchanged.

Definition: According to [Ra68₂], a finite sequence of moves of type (26) – where j may vary – applied to a finite presentation $\mathcal{P} = \langle a_1, \dots, a_g | R_1, \dots, R_h \rangle$ is called a Q -transformation. We speak of Q^* -transformations if, in addition, the following elementary operations are admitted:

- (27) in the relators one replaces a_i throughout by
 a) a_i^{-1} , or b) $a_i a_k$ or c) $a_k a_i$, $k \neq i$.
 (An elementary operation consists of exactly one of the cases a) b) c).
 The generators with index $\neq i$ remain unchanged.)

The operations of type (27) generate all *free* (= Nielsen) transformations¹⁷; see [Ni19].

Q^{**} -transformations¹⁸ are obtained if, moreover, we allow

- (28) to enlarge $\mathcal{P}$ by a new generator a and a new relator $R = a$ (*prolongation*), or to perform the inverse operation if the generator a appears in no other relator than $R = a$.

The Q^{**} -class of $\mathcal{P}$ is also called the *presentation class* of $\mathcal{P}$ and will be denoted by $\Phi(\mathcal{P})$.

Because of the remark leading to (25),

- (29) a finite, connected CW -complex K^2 determines a well defined *presentation class* $\Phi(K^2)$ of $\pi = \pi_1(|K^2|)$,

given by $\Phi(K^2) = \Phi(\mathcal{P})$ for any $\mathcal{P}$ which is read off from K^2 according to § 1.3.

But there is an even stronger connection between Q^{**} -classes and geometry. It is motivated by the (generalized) Andrews-Curtis problem which asks if a simple-homotopy equivalence between finite 2-dimensional CW -complexes K^2, L^2 can always be turned into a 3-deformation, i.e.,

- (30) whether $K^2 \searrow \swarrow L^2$ implies $K^2 \xrightarrow{3} L^2$; see § 4.1 below.

¹⁷In fact, c) and *permutations* of the a_i can be effected by a) and b) (Exercise); inversion and multiplication on one side likewise generate all transformations of the relators in (26) c).

¹⁸Instead of Q^{**} -transformations ([Me76] and [Me79₁]), [AnCu66] and [Cr79₁] use the terminology *extended Nielsen transformations*.

The central result of this subsection is that the *3-deformation types* of compact, connected 2-complexes bijectively correspond to presentation classes:

Theorem 2.4 *The assignment $K^2 \rightarrow \Phi(K^2)$ induces a bijection between 3-deformation types of compact, connected 2-dimensional CW-complexes and Q^{**} -classes of finite presentations, the inverse bijection being induced by associating to a finite presentation $\mathcal{P}$ its standard complex $K_{\mathcal{P}}$.*

Remark: For the p.l. version of this result, see P. Wright [Wr75]; compare also [Yo76]. In the CW-case we will have to deal with technical complications arising from general attaching maps (see the proof of Lemma 2.6 below).

Proof of Theorem 2.4: A) We first consider the transition $\mathcal{P} \rightarrow K_{\mathcal{P}}$. Note that the operations (27) and (28) together allow

- (31) to enlarge $\mathcal{P}$ by a new generator a and a new relator $R = w^{-1}a$ (*generalized prolongation*), where w is a word in the original $a_i^{\pm 1}$, or to perform the inverse operation (if possible).

Moreover, free transformations of the generators (27) can be replaced by α) generalized prolongations and their inverses, and β) Q-transformations (Exercise). Thus we must only check the effect of α) and β) on the associated standard complexes $K_{\mathcal{P}}, K_{\mathcal{P}'}$:

α) A generalized prolongation induces a 2-expansion from $K_{\mathcal{P}}$ to $K_{\mathcal{P}'}$, its inverse a 2-collapse.

β) An elementary Q-transformation replaces the attaching map of e_j^2 (associated to R_j in $K_{\mathcal{P}}$) by one for the 2-cell $e_j'^2$ corresponding to the result of (26) a) b) or c). In each case, the attaching maps of e_j^2 and $e_j'^2$ are homotopic maps from S^1 to $K_{\mathcal{P}} - \{e_j^2\} = K_{\mathcal{P}'} - \{e_j'^2\}$. Hence, by

§ 2.1., Lemma 2.1, there is a transient 3-deformation $K_{\mathcal{P}} \xrightarrow{3} K_{\mathcal{P}'}$.

With respect to later considerations, we note that in this transition from presentation classes to 3-deformation types,

- (32) all complexes, including the 3-dimensional intermediate stages of β), can be assumed to be *PLCW*.

B) The direction $K^2 \rightarrow \Phi(K^2)$ will use two lemmas, the proofs of which we postpone till after (34). The first one and its proof is a generalization of

([Wr75], Lemma 2) to *CW*-complexes and arbitrary dimensions¹⁹:

Lemma 2.5 *If a CW-complex K is transformed into L by expanding finitely many $(n+1)$ -cells e_i^{n+1} (and their free faces) and afterwards collapsing these e_i^{n+1} again (from potentially different free faces), then there is a transformation from K to L by finitely many transient $(n+1)$ -moves.*

This result will help to avoid an accumulation of 3-cells; see A) β).

Lemma 2.6 *If a connected CW-complex contains a 3-cell e^3 with two different free faces e^2, e'^2 , then the process of reading off a presentation for π_1 yields relators R for e^2, R' for e'^2 which satisfy*²⁰

$$(33) \quad R = wR'^{\pm 1}w^{-1} \cdot S,$$

where S is a consequence of those relators which correspond to 2-cells in $\partial e^3 - (e^2 \cup e'^2)$.

Continuing the proof of Theorem 2.4, let K^2 and L^2 be related by a 3-deformation. Because of (14), we may assume $K^2 \nearrow K_1^2, L_1^2 \searrow L'^2$ where K_1^2 and L_1^2 fulfill the hypothesis of Lemma 2.5 with $n = 2$, and L'^2 is isomorphic to L^2 . Hence, there is a deformation $K^2 \nearrow K_1^2 \bigwedge^3 \searrow L_1^2 \searrow L'^2$, the middle symbol indicating a sequence of transient 3-moves. By (25), we may choose any spanning tree etc. when reading off presentation classes. Thus in $K^2 \nearrow K_1^2$, an elementary 1-expansion may be taken to solely enlarge the tree; for a 2-expansion the tree may be kept and the presentation may be enlarged by a generalized prolongation.

Hence, we get $\Phi(K^2) = \Phi(K_1^2)$ and likewise $\Phi(L_1^2) = \Phi(L'^2) = \Phi(L^2)$, the last equality trivially holding because of the isomorphism between L'^2 and L^2 . We finish the argument by showing $\Phi(K_1^2) = \Phi(L_1^2)$: If, in the chain of transient 3-deformations from K_1^2 to L_1^2 , a 3-cell e_j^3 is collapsed from the same free face that was used for its expansion, then the 2-complex hasn't changed during this deformation; if the two free faces are different, then (33) ensures that the corresponding relators before and afterwards differ by a Q-transformation.

¹⁹An n -dimensional and simplicial version of Lemma 2.5 and Theorem 2.4 is given in [KrMe83]. In our case $n = 2$, Lemma 2.5 can be replaced also by an algebraic argument using Lemma 2.6; see [DeMe88].

²⁰By (25) above, an identity of type (33) is independent of the choice of geometric data in § 1.3.

C) If $\mathcal{P}$ is read off from K^2 , then the composed transitions $K^2 \rightarrow \mathcal{P} \rightarrow K_{\mathcal{P}}$ give rise to a 3-deformation: The contraction of a spanning tree of K^2 can be achieved by a 3-deformation (see (15) above) without changing the R_j to be read off; the 2-cell corresponding to R_j can then be normalized to the one of $K_{\mathcal{P}}$ by an application of Lemma 2.1.

Once more we observe that

(34) if K^2 is a *PLCW*-complex, then $K^2 \xrightarrow{3} K_{\mathcal{P}}$ can be achieved by a *PLCW*-3-deformation.

For the opposite direction of compositions, it is obvious that the data in reading off a presentation from $K_{\mathcal{P}}$ can be chosen such that $\mathcal{P} \rightarrow K_{\mathcal{P}} \rightarrow \mathcal{P}$.

Hence the assignments $K^2 \rightarrow \Phi(K^2)$ and $\mathcal{P} \rightarrow K_{\mathcal{P}}$ induce inverse maps between 3-deformation types of compact connected 2-dimensional *CW*-complexes and Q^{**} -classes of finite presentations. $\square$

Remarks: Using (32) and (34), it follows 1) that Theorem 2.4 is also true in the *PLCW*-case; 2) (via the transition to presentation classes:) *PLCW*-complexes K^2 and L^2 which fulfill $K^2 \xrightarrow{3} L^2$ as *CW*-complexes are already related by a *PLCW*-3-deformation (compare (18) above). The latter holds iff there exists a p.l. 3-deformation from $|K^2|$ to $|L^2|$; see [Wr75].

Proof of Lemma 2.5: Let D be the given deformation from K to L . Enumerate the $(n+1)$ -cells $e_1^{n+1}, \dots, e_k^{n+1}$ in the order in which they appear in D , and let e_i^n denote the free face through which e_i^{n+1} is eventually collapsed. Construct a transient deformation D' from K to L in the following manner: When e_1^{n+1} is expanded in D , let it also be expanded in D' but immediately collapsed via e_1^n . Instead of expanding e_2^{n+1} in D , we expand $e_2'^{n+1}$ (in D'), which is obtained from e_2^{n+1} by identifying²¹ each point of $\partial e_2^{n+1} \cap e_1^n$ with its image under a retraction map $\partial e_1^{n+1} \rightarrow (\partial e_1^{n+1}) - (e_1^n)$. e_2^n must be a free face of $e_2'^{n+1}$: this could fail only if, in D both $e_1^n \cap \partial e_2^{n+1}$ and $e_2^n \cap \partial e_1^{n+1}$ were nonempty, which is impossible since it would block the collapse of e_1^{n+1} and e_2^{n+1} in D . Collapse $e_2'^{n+1}$ via e_2^n .

Similarly, replace each subsequent e_i^{n+1} of D by $e_i'^{n+1}$ in D' , which is obtained from e_i^{n+1} by dragging each point of ∂e_i^{n+1} along previous retraction maps of D' . $e_i'^{n+1}$ can be collapsed immediately via e_i^n , for otherwise there would exist some “cycle” of nonempty intersections: $e_i^n \cap \partial e_{i_1}^{n+1} \neq \emptyset$, $e_{i_1}^n \cap \partial e_{i_2}^{n+1} \neq \emptyset, \dots, e_{i_m}^n \cap \partial e_i^{n+1} \neq \emptyset$ with disjoint indices; and D could never have entered the cells $e_i^{n+1}, e_{i_1}^{n+1}, \dots, e_{i_m}^{n+1}$. $\square$

²¹Similar quotients turn attaching/characteristic maps for e_2^{n+1} into analogous maps for (a well defined) $e_2'^{n+1}$, although different ones might be used for the expansion and the collapse; see footnote 22.

Proof of Lemma 2.6: By the freeness of e^2 and e'^2 , we have characteristic maps $\Phi, \Phi' : I^3 \rightarrow \bar{e}^3$ such that

$$(35) \quad \begin{aligned} \Phi|I^2 \times \{0\} \text{ is characteristic for } e^2, \quad \Phi^{-1}(\bar{e}^2) &= I^2 \times \{0\}; \\ \Phi'|I^2 \times \{1\} \text{ is characteristic for } e'^2, \quad \Phi'^{-1}(\bar{e}'^2) &= I^2 \times \{1\}. \end{aligned}$$

It would be easy to prove (33) if we could assume in addition that Φ equals Φ' . But in general $\Phi|I^2 \times \{1\}$ and $\Phi'|I^2 \times \{0\}$ will not be injective. We use homotopy considerations²² to bypass the problem of finding a characteristic map for e^3 which simultaneously establishes e^2 and e'^2 as free:

Select a point p of e'^2 . Although $\Phi^{-1}(p)$ in general will be different from a single point, it is not hard to see that

$$(36) \quad \Phi^{-1}(p) \text{ is a compact, connected subset of } \partial I^3 - (I^2 \times \{0\}).$$

(Compare Φ and Φ' , details left as an exercise.)

$\Phi^{-1}(p)$ then is contained in the interior $\overset{\circ}{M}$ of a compact, connected(!) p.l. submanifold M^2 of $\partial I^3 - (I^2 \times \{0\})$, which fulfills $\Phi(M^2) \subseteq e'^2$.

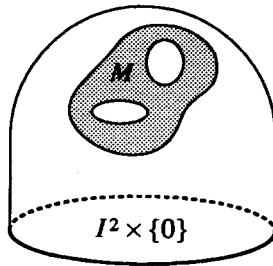


Figure I.18.

As $\Phi|(\partial I^3 - \overset{\circ}{M}) : (\partial I^3 - \overset{\circ}{M}) \rightarrow \partial e^3$ does not meet p , we can deform this map rel. $I^2 \times \{0\}$ to miss e^2 entirely. This homotopy extends to a homotopy of $\Phi|\partial I^3$, which maps $M^2 \times I$ to $\bar{e}^2$. Applying cellular approximation rel. $I^2 \times \{0\}$ ([Schu64], III. 3.5), we get an $f : \partial I^3 \rightarrow \partial e^3$ such that

$$(37) \quad \text{a) } f \simeq \Phi|\partial I^3 \text{ rel. } I^2 \times \{0\};$$

²²The proof given here has been communicated privately since 1976; see ([Me79₁], footnote 2). R.A.Brown [Br92] established the existence of simultaneous characteristic maps for free faces of CW-3-cells. The analogous question for faces of n -cells, $n > 3$, seems to be open.

- b) $f^{-1}(e^2) = \overset{\circ}{I}^2 \times \{0\}$ (this property can be preserved throughout the deformation);
- c) $f^{-1}(e'^2) \subseteq M^2$ and $f(M^2) \subseteq \bar{e}'^2$;
- d) f maps the 1-skeleton (according to Figure 19) of a CW -decomposition of ∂I^3 to the portion of ∂e^3 that lies in the 1-skeleton.

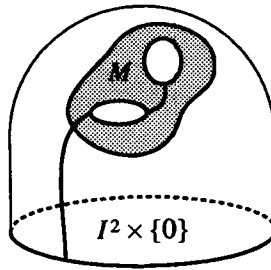


Figure I.19.

As all of ∂M is mapped to $\partial e'^2$ by f , this cellular dissection of ∂I^3 and (37) give rise to an identity of the form²³

$$(38) \quad R = wR'w^{-1} \cdot S, \quad \gamma \in \mathbb{Z},$$

with R, R', S having the same meaning as in (33). (In short: The connectivity of $\Phi^{-1}(p)$ and of M yields that there is only *one* “ R' -patch” in (38).) But f is homotopic to $\Phi|\partial I^3$; hence, γ is the incidence number $[e^3, e'^2]$, which is independent of the characteristic map for e^3 ; compare Chapter II, § 3. Using Φ' , i.e., the freeness of e'^2 , we have $\gamma = [e^3, e'^2] = \pm 1$. $\square$

Theorem 2.7 *The presentation class $\Phi(K^2)$ is an invariant of the homeomorphism type²⁴ of $|K^2|$.*

Proof: Let K^2, L^2 be finite, connected CW -complexes which are homeomorphic. Without loss of generality, we may assume $|K| = |L|$. Select a finite number of points x_i such that each 2-cell of K^2 and each 2-cell of L^2 contains at least one x_i , and none of the x_i lies in $|K^1| \cup |L^1|$. There also exist pairwise disjoint 2-disc-neighbourhoods $U(x_i)$ which still don't meet the 1-skeleta. Define $\mathcal{J}$ to be $|K| - \bigcup_i \overset{\circ}{U}(x_i) = |L| - \bigcup_i \overset{\circ}{U}(x_i)$.

²³Compare footnote 20.

²⁴Compare (23) above. In the p.l. case, a proof of Theorem 2.7 could be based on the Hauptvermutung; see § 1.2. But with respect to Theorem 1.2 of § 1.2, the CW -case also has to take care of combinatorially distinct 2-complexes.

Every 2-cell is punctured at least once; hence $\pi_1(\mathcal{J})$ is free (see below for a concrete basis). We will show that

(39) by expressing the boundaries $\partial U(x_i)$ ($\triangleq$ defining relators of a presentation) with respect to any²⁵ basis ($\triangleq$ generators of the presentation) of $\pi_1(\mathcal{J})$, we obtain a member of $\Phi(K^2)$.

As (39) symmetrically holds for $\Phi(L^2)$ as well as for $\Phi(K^2)$, this yields $\Phi(K^2) = \Phi(L^2)$. Thus it suffices to prove (39):

Reindex the $U(x_i)$ such that e_j^2 contains $U(x_{j,1}, \dots, U(x_{j,k_j}))$, $k_j \geq 1$. Subdivide K^2 by connecting the $\partial U(x_{j,\nu})$ with ∂e_j^2 according to Figure 20. This figure also indicates further obvious notations.

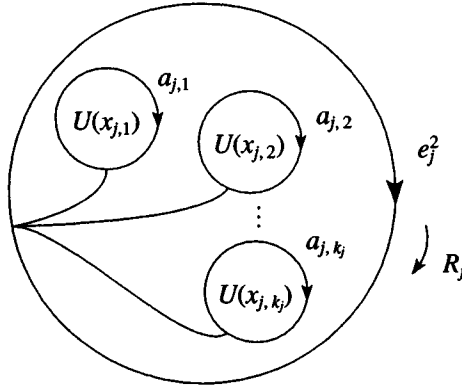


Figure I.20.

Note that $\pi_1(\mathcal{J})$ is freely generated by a basis of $\pi_1(K^1)$ together with the $a_{j,\nu}$, $\nu \geq 2$. (To see this, collapse the $e_j^2 - \bigcup_{\nu=1}^{k_j} U(x_{j,\nu})$ from $\partial U(x_{j,1})$). With respect to this basis the $\partial U(x_{j,\nu})$ give rise to the relators $a_{j,1} = R_j \cdot a_{j,k_j}^{-1} \cdot \dots \cdot a_{j,2}^{-1}$, $a_{j,2}, \dots, a_{j,k_j}$. Hence we have to show that the presentation $\mathcal{P} = \langle \pi_1(K^1) - \text{basis}, a_{j,\nu} (\nu \geq 2) | R_j \cdot a_{j,k_1}^{-1} \cdot \dots \cdot a_{j,2}^{-1}, a_{j,2}, \dots, a_{j,k_j} \rangle$ (with j ranging over the 2-cells of K^2) is Q^{**} -equivalent to (the) one read off from K^2 . But this follows from the chain of Q^{**} -transformations

$$\begin{aligned} \mathcal{P} &\rightarrow \langle \pi_1(K^1) - \text{basis}, a_{j,\nu} (\nu \geq 1) | R_j \cdot a_{j,k_1}^{-1} \cdot \dots \cdot a_{j,2}^{-1}, a_{j,1}, \dots, a_{j,k_j} \rangle \\ &\rightarrow \langle \pi_1(K^1) - \text{basis}, a_{j,\nu} (\nu \geq 1) | R_j, a_{j,1}, \dots, a_{j,k_j} \rangle \rightarrow \langle \pi_1(K^1) - \text{basis} | R_j \rangle. \quad \square \end{aligned}$$

²⁵ A different one would result in a Q^{**} -transformation of type (27).

Discussion and Additions:

If one enlarges Q^{**} -transformations by the elementary operation which allows to pass from $\mathcal{P} = \langle a_1, \dots, a_g | R_1, \dots, R_h \rangle$ to $\mathcal{P} = \langle a_1, \dots, a_g | R_1, \dots, R_h, 1 \rangle$, or to perform the inverse operation (if possible), then all²⁶ *Tietze transformations* are obtained. By these it is possible to convert any two (finite) presentations of a given group π into each other (Exercise, *Theorem of Tietze* [Ti08]). Q^{**} -transformations, however, preserve the *deficiency*²⁷ $h - g$ of $\mathcal{P}$, or, equivalently, 3-deformations preserve the Euler characteristic $\chi(|K^2|) = 1 - g + h$. The theorem of Tietze yields that

- (40) any two finite, connected CW -complexes K^2 and L^2 with isomorphic fundamental groups fulfill²⁸ $K^2 \vee S_1^2 \vee \dots \vee S_k^2 \xrightarrow{3} L^2 \vee S_1^2 \vee \dots \vee S_\ell^2$ for suitably chosen finite numbers k, ℓ of 2-spheres.

But until the mid seventies it was unknown, whether (a finitely presentable group) π and the deficiency together are sufficient to determine a presentation class or not: Although an algebraic homotopy classification of finite, connected 2-complexes had been obtained (see Chapter II, § 4), one didn't have examples with $K^2 \not\cong L^2$ but equal π_1 and χ . Such examples were first given by Dunwoody [Du76] and Metzler [Me76]; see also Sieradski [Si77]. A systematic treatment of this phenomenon is a main topic of Chapter III. Presentation classes even contain genuine simple-homotopy information, as simple-homotopy type and homotopy type differ already in dimension 2, see the references given in (21) above. At the time of writing, only (30) contributes a question mark in the following chain of implications:

(41)

$$K^2 \xrightarrow{3} L^2 \stackrel{?}{\Longleftrightarrow} K^2 \frown L^2 \stackrel{?}{\Longleftrightarrow} K^2 \simeq L^2 \stackrel{?}{\Longleftrightarrow}$$

$$\pi_1(|K^2|) \approx \pi_1(|L^2|) \text{ and } \chi(|K^2|) = \chi(|L^2|),$$

where K^2, L^2 are finite, connected CW -complexes.

Because of (40), all distinguishing examples necessarily have a “small” Euler characteristic; in many cases the minimal value of χ is required; see Chapters III, VII and IX.

²⁶The usual list of Tietze transformations (generalized prolongation, adding an arbitrary consequence of the R_j , and the inverses of these operations) is easily seen to be equivalent to this larger one; see the exercise after (31).

²⁷Sometimes the deficiency is defined to be $g - h$.

²⁸From this it can be deduced that there is no algorithm which decides whether two arbitrary finite presentations are Q^{**} -equivalent or not; see [Me85] and Chapter XII, § 2.1.

Every finite, connected *CW*-complex K^3 with $K^3 \searrow K^2$ determines a presentation class by defining $\Phi(K^3) = \Phi(K^2)$. It is well defined for, if L^2 is another 2-spine of K^3 , then $K^2 \nearrow K^3 \searrow L^2$ implies $\Phi(K^2) = \Phi(L^2)$ (Theorem 2.4). The analogous definition can be made in the p.l. case (see the remarks following the proof of Theorem 2.4).

In particular, *every compact, connected (p.l.) 3-manifold M^3 with nonempty boundary determines a presentation class $\Phi(M^3)$* . If $\partial M = \emptyset$, we first remove the interior $\overset{\circ}{D}^3$ of a p.l. 3-ball D^3 in M^3 and define $\Phi(M^3)$ to be the presentation class of the complement. By connectedness this doesn't depend on the choice of D^3 ([He76], Th. 1.5) and, because of the Hauptvermutung for 3-manifolds, $\Phi(M^3)$ is a *topological invariant*. It may be considered as the “algebraic essence” which remains when one “forgets” that relators live on a surface (compare § 3.1 as well as Reidemeister [Re33] and Singer [Si33]).

Clearly

$$(42) \quad \Phi(S^3) = \Phi(D^3) = \Phi_0,$$

where Φ_0 is the class of the empty presentation or of a *trivial presentation* $\langle a_1, \dots, a_g | a_1, \dots, a_g \rangle$.

Presentation classes Φ and Ψ may be added by taking “disjoint” representatives $\langle a_i | R_j \rangle$, $\langle b_k | S_\ell \rangle$ and passing to the class of $\langle a_i, b_k | R_j, S_\ell \rangle$. The sum corresponds to the one-point union of 2-complexes and the free product of the groups which are presented; see Chapter XII, § 3. Thus one gets an *abelian semigroup $\mathcal{H}$ of presentation classes* with the neutral element Φ_0 . But several interesting algebraic questions on this semigroup are still unsolved; see § 4.4.

3 P.L. Embeddings of 2-Complexes into Manifolds

If K is a subcomplex in the interior of a triangulated p.l. manifold M , then a sufficiently small “regular” p.l. neighbourhood $N(|K|)$ of K in M may be obtained as follows: Thicken the 0-simplices to p.l. balls, and connect those by “tubes”, for which there is a connecting 1-simplex in K ; see Figure 21.

The result is a “handlebody around” K^1 . Continue to thicken, if K contains simplices of dimensions ≥ 2 ; see again Figure 21: they likewise give rise to handles of corresponding *indices* ≥ 2 which are attached inductively to the boundary of the previous thickening.